

MUHAMMAD RAKEH SALEEM *Ph.D.*

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A highly driven **Data Scientist and Machine Learning Engineer** with a Ph.D. and over 5 years of experience leading and implementing end-to-end ML solutions in computer vision and deep learning. Expertise spans the full ML model lifecycle: **data engineering, model optimization, deployment, and systems development**. Proven track record in leveraging image processing, object detection, and deep neural networks to translate complex research into scalable, high-impact production code. Seeking a Data Scientist role to drive Digital Transformation by applying advanced data science and AI/ML capabilities.

Education

Pennsylvania State University

Aug 2020 – Dec 2024

PhD in Engineering

State College, USA

- **Thesis:** From Gaze to Insight: Leveraging Eye Tracking for Structural Assessment and Evaluation

Chung-Ang University

Sept 2018 – Jun 2020

Master in Engineering

Seoul, S. Korea

- **Thesis:** Bridge Inspection Using an Unmanned Aerial Vehicle by Image Capturing and Geo-tagging System and Deep Convolutional Neural Network

Bahria University Islamabad

Aug 2013 – Jun 2017

Bachelor in Electrical Engineering

Islamabad, Pakistan

- **Thesis:** FPGA-based Digital Signal Processing of Under-sea Range Finder

Research Experience

Pennsylvania State University

Sep 2020 – Dec 2024

Researcher – Built Environment Analytics and Modeling Lab

State College, USA

- Led a multi-year research agenda to build and deploy complex **computer vision and deep learning solutions**, demonstrating technical leadership and full-stack ML ownership.
- Developed and optimized CNN models for high-resolution image analysis, achieving **95% accuracy** in object detection while **reducing inference time by 40%** via model optimization and efficient serving layers.
- Designed and implemented **machine learning pipelines** for production use, leveraging user perception data (eye-tracking) to inform model training and novel evaluation techniques for human-AI systems.
- Developed algorithms for **visual search** and **image classification/processing** that translated human expert knowledge into quantifiable ML features for autonomous navigation systems.

Chung-Ang University

Sep 2018 – Aug 2020

Research Assistant – Smart Infrastructure and Technology Lab

Seoul, S. Korea

- Developed and deployed deep convolutional neural networks (**deep CNNs**) for real-time visual analysis, including a system for construction worker safety monitoring that **achieved a 25% improvement** in risk detection speed.
- Designed the end-to-end development of an integrated UAV imaging system, involving **hardware-software integration** and **image stitching algorithms** to render high-resolution data from 50 inspection flights.
- Focused on applied ML, implementing **CNN models on-device** to demonstrate a full-stack, embedded programming solution from data acquisition to intelligent analysis, a key skill for production systems.

Work Experience

Pennsylvania State University

Sep 2021 – Aug 2023

Teaching Assistant— EDSGN 100: Cornerstone Engineering Design

State College, USA

- Developed and delivered technical lectures and practical sessions on complex engineering and computational methods to undergraduate students.
- Fostered a collaborative learning environment and demonstrated strong communication and mentorship skills by guiding students through intricate design and computational challenges.

- Led the design and development of an **end-to-end IoT system**, from embedded programming to software framework and cloud-based data visualization within a 3-month timeline.
- Engineered a wireless sensor system for fault identification and localization, implementing an embedded programming solution on a microcontroller that **achieved 97% data transmission reliability**.
- Built a proof-of-concept for a **real-time data analysis and visualization dashboard** on the ThingSpeak Cloud platform, showcasing experience with data pipelines and back-end systems, and UX layers.

Software & Technical Skills

- **Programming:** Python, C++, MATLAB, SQL
- **ML Infrastructure:** Model Deployment, Optimization, Evaluation, Data Pipelines, Debugging
- **Computer Vision:** Object Detection, Image Processing, Image Classification, Visual Search, Photogrammetry
- **Software/Systems:** Data Structures, Algorithms, System Diagnosis, Embedded Systems, IoT
- **ML/AI:** Deep Learning, CNNs, TensorFlow, Keras, scikit-learn
- **Tools/Platforms:** Git, Linux/Unix, Jupyter Notebook, Cloud Platforms

Invited Talks

- 2024 **Thornton Tomasetti**, “From gaze to insight: Leveraging eye tracking for SHM”
- 2023 **ETH Zurich**, “AI-enabled damage detection using eye tracking”
- 2020 **PennState**, “Digital twins and AI-enabled SHM”
- 2020 **PennState**, “Does more data lead to more answers or questions?”

Publications

- **M.R. Saleem** and R. Napolitano. Comparative Analysis of Saliency Map Algorithms in Capturing Visual Priorities for Building Inspections, **Journal of Building Engineering**, 97, 1106, 2024.
- **M.R. Saleem**, R. Mayne, and R. Napolitano. Evaluating Human Expert Knowledge In Damage Assessment Using Eye Tracking: A Disaster Case Study, **Buildings**, 14(7), 2114, 2024.
- **M.R. Saleem**, R. Mayne, and R. Napolitano. Analysis of gaze patterns during facade inspection to understand inspector sense-making processes. **Scientific Reports**, 13(1), 2929, 2023.
- N. Khan, **M.R. Saleem**, D. Lee, M.W. Park, and C. Park. Utilizing safety rule correlation for mobile scaffolds monitoring leveraging deep convolution neural networks. *Computers in Industry*, 129, 2021.
- **M.R. Saleem**, J.W. Park, J.H. Lee, H.J. Jung, and M.Z. Sarwar. Instant bridge visual inspection using an unmanned aerial vehicle by image capturing and geo-tagging system and deep convolutional neural network, **Structural Health Monitoring**, 20(4), 1760-1777, 2021.

Mentoring

- **Led the mentorship of 8 undergraduate and graduate students** on research projects spanning eye-tracking, image processing, and computer vision, resulting in multiple publications and conference presentations.